

1     **Supplementary material for “Fusing short-pulse voltage waveforms**  
2         **with handcrafted features for rapid state-of-health estimation of**  
3     **lithium iron phosphate batteries: a controlled comparison of encoder**  
4             **pairings”**

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## 15 **Supplementary Note S1. Handcrafted feature definitions**

16 This note defines the 143 waveform-derived features of Table 2; the seven  
 17 context quantities are given in Eq. (3) of the main text. The segments follow the  
 18 publisher's pulse timing and are defined, together with the first and second  
 19 differences, as

$$20 \quad S^{(1)} = (V_1, \dots, V_{30}), S^{(2)} = (V_{31}, \dots, V_{40}), S^{(3)} = (V_{41}, \dots, V_{100}), d_t^{(1)} = V_{t+1} - V_t, d_t^{(2)} = d_{t+1}^{(1)} - d_t^{(1)} \\ 21 \quad \quad \quad (S1)$$

23 **Basic operators.** For a window  $x$  of length  $n$ ,

$$24 \quad \bar{x} = \frac{1}{n} \sum_{i=1}^n x_i, \sigma(x) = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2}, R M S(x) = \sqrt{\frac{1}{n} \sum_{i=1}^n x_i^2}, E(x) = \sum_{i=1}^n x_i^2 \quad (S2)$$

$$25 \quad R(x) = \max x - \min x, I Q R(x) = Q_{0.75} - Q_{0.25}, \\ M A D(x) = \text{median}(|x - \text{median}(x)|), A U C(x) = \sum_{i=1}^{n-1} \frac{x_i + x_{i+1}}{2} \quad (S3)$$

26 The standard deviation uses the population form  $(1/n)$ , except for the global  
 27 voltage sequence, which follows the publisher's supplementary definition with the  
 28 sample form  $(1/(n-1))$ ; both are labelled in the feature list.

29 **Shape.** Bias-corrected Fisher-Pearson skewness and excess kurtosis are used,  
 30 with  $m_k$  the  $k$ -th central moment:

$$31 \quad m_k = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^k, g_1 = \frac{m_3}{m_2^{3/2}}, g_2 = \frac{m_4}{m_2^2} - 3, \\ G_1 = \frac{\sqrt{n(n-1)}}{n-2} g_1, G_2 = \frac{n-1}{(n-2)(n-3)} [(n+1)g_2 + 6] \quad (S4)$$

32 **Histogram entropy.** The range of the window is divided into 16 equal bins and  
 33  $p_i$  is the fraction of samples in bin  $i$ :

$$34 \quad H(x) = -\frac{1}{\ln 16} \sum_{p_i > 0} p_i \ln p_i \quad (S5)$$

35 **Linear trend.** A least-squares line is fitted against the sample index  $t_i$ :

$$a = \frac{\sum_i (x_i - \bar{x})(t_i - \bar{t})}{\sum_i (t_i - \bar{t})^2}, b = \bar{x} - a\bar{t}, R^2 = 1 - \frac{\sum_i (x_i - b - at_i)^2}{\sum_i (x_i - \bar{x})^2} \quad (S6)$$

**Zero crossings and total variation.** Zero crossings of a difference sequence are counted after removing exact zeros, giving  $y$ :

$$Z(y) = \#\{i: y_i y_{i+1} < 0\}, TV(x) = \sum_{i=1}^{n-1} |x_{i+1} - x_i| \quad (S7)$$

The 15 operators of Eqs. (S2)–(S7) applied to  $S^{(1)}$ ,  $S^{(2)}$  and  $S^{(3)}$  give the segment statistics; applied to the full sequence and to  $d^{(1)}$  and  $d^{(2)}$ , together with end points and quantiles, they give the 24 global and 22 difference-dynamics features.

**Spectral features.** The demeaned sequence is Hann-windowed and transformed by a real fast Fourier transform ( $N=101$ , frequency in cycles per sample):

$$w_t = 0.5 - 0.5 \cos\left(\frac{2\pi t}{N-1}\right), x_t = (V_t - \bar{V}) w_t, \quad (S8)$$

$$X_k = \sum_{t=0}^{N-1} x_t e^{-j2\pi kt/N}, P_k = |X_k|^2, f_k = \frac{k}{N}$$

The amplitudes of the first eight non-DC lines and the energy shares of four bands are

$$A_k = \frac{2|X_k|}{\sum_t w_t} (k=1, \dots, 8), \quad (S9)$$

$$B_1 = \frac{\sum_{k=1}^3 P_k}{\sum_{k=1}^{50} P_k}, B_2 = \frac{\sum_{k=4}^7 P_k}{\sum_{k=1}^{50} P_k}, B_3 = \frac{\sum_{k=8}^{15} P_k}{\sum_{k=1}^{50} P_k}, B_4 = \frac{\sum_{k=16}^{50} P_k}{\sum_{k=1}^{50} P_k}$$

Spectral-shape features exclude the DC component: dominant frequency and amplitude, centroid, spread, flatness, entropy and 85% roll-off,

$$k^* = \arg \max_{k \geq 1} P_k, f_{dom} = f_{k^*}, A_{dom} = A_{k^*}, \quad (S10)$$

$$C = \frac{\sum_k f_k P_k}{\sum_k P_k}, S = \sqrt{\frac{\sum_k P_k (f_k - C)^2}{\sum_k P_k}}, F = \frac{\left(\prod_k P_k\right)^{1/K}}{\frac{1}{K} \sum_k P_k}$$

$$H_f = -\frac{1}{\ln 50} \sum_k p_k \ln p_k, p_k = \frac{P_k}{\sum_{\kappa} P_{\kappa}}, f_{85} = \min \left\{ f_k : \sum_{\kappa \leq k} P_{\kappa} \geq 0.85 \sum_{\kappa} P_{\kappa} \right\} \quad (S11)$$

**Wavelet features.** A four-level db2 decomposition with symmetric extension gives coefficient sets  $c$ , and their energy fractions are

$$\varepsilon_c = \sum_i c_i^2, EF_c = \frac{\varepsilon_c}{\sum_{c'} \varepsilon_{c'}}, c \in \{cA_4, cD_4, cD_3, cD_2, cD_1\} \quad (S12)$$

Eqs. (S8)–(S12) yield 24 frequency and wavelet features (8 amplitudes, 4 band shares, 7 spectral-shape features and 5 wavelet energy fractions). The dominant frequency takes a single value in the whole dataset and the roll-off frequency takes two values and is constant in the training cells of fold 0; their rank correlations are undefined in the affected folds and they are skipped by the finiteness condition of Section 2.3.3, which is why the feature-count ladder ends at 144 rather than 146.

**Nonlinear features.** Lagged autocorrelation ( $\ell \in \{1, 5, 10\}$ ) and the three Hjorth parameters are

$$a = (V_0, \dots, V_{100-\ell}), b = (V_{\ell}, \dots, V_{100}), \rho_{\ell} = \frac{\sum_i (a_i - \bar{a})(b_i - \bar{b})}{\sqrt{\sum_i (a_i - \bar{a})^2 \sum_i (b_i - \bar{b})^2}}, \quad (S13)$$

$$Act = \sigma^2(V), Mob(V) = \sqrt{\frac{\sigma^2(d^{(1)})}{\sigma^2(V)}}, Com = \frac{Mob(d^{(1)})}{Mob(V)}$$

Approximate and sample entropy use embedding dimension  $m=2$  (and  $m+1=3$ ), tolerance  $r=0.2\sigma(V)$  and the Chebyshev distance; approximate entropy counts self-matches, whereas sample entropy counts only pairs  $i < j$ :

$$C_i^m(r) = \frac{\#\{j: \max_{0 \leq k < m} |V_{i+k} - V_{j+k}| \leq r\}}{N - m + 1}, \Phi^m(r) = \frac{1}{N - m + 1} \sum_i \ln C_i^m(r), \quad (S14)$$

$$ApEn = \Phi^2(r) - \Phi^3(r), SampEn = -\ln \frac{A}{B}$$

Permutation entropy uses order 3 and delay 1, and the Hurst exponent is the slope of a variogram regression over lags  $\ell \in \{1, 2, 4, 8, 16\}$ :

$$PE = -\frac{1}{\ln 6} \sum_{\pi \in \Pi_3} p(\pi) \ln p(\pi), \tau(\ell) = \sqrt{\frac{1}{101-\ell} \sum_i (V_{i+\ell} - V_i)^2}, \quad (S15)$$

$H = \text{OLS slope of } \ln \tau(\ell) \text{ on } \ln \ell$

**Transient-boundary features.** Fifteen voltage differences between specified samples, including the steps at current changes ( $V_1 - V_0, V_{31} - V_{30}, V_{41} - V_{40}$ ) and the net changes within segments ( $V_{30} - V_1, V_{40} - V_{31}, V_{100} - V_{41}$ ), describe the transitions between segments. Three response times are defined on the rest segment  $S^{(3)}$ , with linear interpolation between samples and  $\alpha \in \{0.10, 0.50, 0.90\}$ :

$$p_t = \text{sgn}(V_{100} - V_{41})(V_t - V_{41}), t = 41, \dots, 100, t_\alpha = \min\{t : p_t \geq \alpha |V_{100} - V_{41}|\} \quad (S16)$$

## Supplementary Note S2. PatchTST encoder equations

Each patch  $p_m$  of Eq. (12) is embedded by a shared linear map and a learnable positional embedding  $e_m$ , initialised from  $N(0, 0.02^2)$  and trained with the network:

$$z_m^{(0)} = W_p p_m + b_p + e_m, W_p \in R^{D \times P}, Z^{(0)} = [z_1^{(0)}, \dots, z_M^{(0)}]^T \in R^{M \times D} \quad (S17)$$

The  $M$  tokens enter one Transformer encoder layer whose multi-head self-attention ( $h=4$  heads,  $d_k=D/h=8$ ) acts between patches:

$$Q_i = Z W_i^Q, K_i = Z W_i^K, V_i = Z W_i^V, \\ head_i = \text{softmax}\left(\frac{Q_i K_i^T}{\sqrt{d_k}}\right) V_i, MHA(Z) = [head_1; \dots; head_h] W^O \quad (S18)$$

The layer uses post-normalisation residual connections and a GELU feed-forward sublayer of width  $2D=64$  with dropout 0.2:

$$Z' = LN(Z^{(0)} + MHA(Z^{(0)})), Z^{(1)} = LN(Z' + FFN(Z')), \\ FFN(u) = W_2 GELU(W_1 u + b_1) + b_2 \quad (S19)$$

Layer normalisation, mean pooling over tokens and a linear projection give the 32-dimensional waveform embedding:

$$E_v(\tilde{v}) = \text{Drop}\left(\text{ReLU}\left(W_o \cdot \frac{1}{M} \sum_{m=1}^M LN(z_m^{(1)}) + b_o\right)\right) \in R^{32} \quad (S20)$$

91 **Supplementary Note S3. Statistical test families**

92 Holm correction was applied within the following prespecified families at  $\alpha =$   
93 0.05: (1) the 14 encoder pairings versus MLP + CNN; (2) the five waveform  
94 complements versus A2; (3) the confirmatory ablations R0 – A1, R0 – A2, R0 – A3  
95 and A3 – A4; (4) the three patch configurations versus R0; (5) the three adaptation  
96 ablations versus R0; (6) the six feature-count levels versus  $K = 50$ , separately for  
97 RMSE and MAE; and (7) the reproduced reference model versus A2 and versus R0  
98 (an extended family whose adjusted  $p$  values are conservative). A1 – A2 was a single  
99 prespecified comparison without correction. Marginal effects, difference-in-  
100 differences, stratified increments, residual complementarity and low-SOH bias are  
101 descriptive. Confidence intervals are conditional on the fixed split and the trained  
102 models and do not cover the variability of new data collection or repartitioning.

103 **Supplementary Note S4. Re-implementation used for the generalisation,**  
104 **robustness and deployment experiments**

105 Apart from the main NMC/graphite results, the experiments of Section 4.6 were  
106 run with an independent re-implementation of the protocol written in Python (NumPy,  
107 scikit-learn 1.9 and PyTorch 2.14 on CPU, one thread per run), starting from the  
108 publisher's processed pulse files [31]. Cleaning reproduced the record counts exactly:  
109 17,130 records in 2,855 groups before cleaning, 192 records removed at the 42nd  
110 RPT, and 16,938 records in 2,823 complete groups (SOH 72.34–100%) after cleaning.  
111 The 143 waveform features were implemented from the definitions in Supplementary  
112 Note S1; details not fixed by those definitions, such as the quantile levels of the global  
113 family and the nine voltage differences not listed in Note S1, may differ from the  
114 original code, and the response-time levels were set to 0.5, 0.632 and 0.9. Feature  
115 selection was nonetheless similar: in fold 0 it selected 16 frequency and wavelet, 14  
116 segment, 11 difference-dynamics, 6 global, 1 boundary and 1 nonlinear feature

together with the DCIR, which ranked 13th; 44 features were selected in every fold and 57 in at least one. The split permutation used NumPy’s default generator with seed 42, which reproduces the fold sizes but not necessarily the fold membership of the original runs. Architectures, loss, optimiser, scheduler, early stopping and seeds follow Sections 3.2–3.4 of the main text, with positional embeddings initialised from  $U(-0.02, 0.02)$ .

Under the original five-fold protocol the re-implementation gives  $R0$  1.427 and  $A2$  1.474 pp (reported: 1.377 and 1.500 pp). The waveform increment keeps its sign and significance ( $-0.046$  pp,  $-0.070$  to  $-0.023$ ;  $p < 0.001$ ; 42 of 64 cells), and cells 12 and 3 are again the best- and worst-estimated cells. Table S6 lists all re-implementation results.

*NMC/graphite dataset.* The UConn-ILCC NMC dataset [31] records the same 100-s pulses at nine nominal SOCs; the three SOCs used for LFP (20%, 50% and 90%) were retained, giving six conditions per RPT, 8,016 records in 1,336 cell–RPT groups of 44 cells and SOH from 41.2% to 100%. Because NMC cells have a higher resistance, the voltage-step and range limits of Eq. (2) were relaxed to 0.3 and 0.8 V; no record was removed. The main-text results come from the original implementation (five folds  $\times$  three seeds,  $K = 50$ ,  $\lambda = 0.1$ , as for LFP):  $R0$  1.968 pp,  $A2$  2.132 pp,  $R0 - A2 = -0.163$  pp (Holm-adjusted  $p = 0.0004$ , 39/44) and  $A1$  2.468 pp. The DCIR correlates strongly with SOH ( $\rho \approx -0.82$ ) but entered none of the 50 selected features because it is redundant with selected waveform features, so it was added to the  $A1$  input separately, as the definition of  $A1$  requires;  $R0$  and  $A2$  were run unchanged.

*Difference between the re-implementation and the original implementation.* The re-implementation found no waveform increment on NMC ( $R0$  2.030 pp,  $A2$  2.026 pp,  $R0 - A2 = +0.004$  pp,  $p = 0.90$ ). Diagnostics trace the difference to the backbone

142 features. The re-implementation computes its features on the absolute voltage and, on  
 143 NMC, selects absolute-level features such as the pre-pulse and final voltages and  
 144 voltage quantiles (six in fold 0, almost none on LFP). Because the NMC open-circuit  
 145 voltage is sloped and capacity fade shifts the actual SOC of each pulse, the pre-pulse  
 146 voltage tracks SOH (mean absolute Spearman correlation 0.93, LFP 0.36) and  
 147 predicts the actual SOC deviation with  $R^2 = 0.964$  (LFP 0.489). Once the backbone  
 148 holds this information, the dynamic information of the waveform branch partly  
 149 overlaps with it: within each pulse condition the selected features explain 99.4% of  
 150 the variance of the first ten principal components of  $\Delta V$  (LFP 18.3%). With the 28  
 151 absolute-level features removed from the candidates, the re-implementation gives A2  
 152 2.098 pp, R0 1.940 pp and  $R0 - A2 = -0.158$  pp ( $-0.205$  to  $-0.112$ ,  $p < 0.001$ , 36/44),  
 153 consistent with the original implementation. Using all nine SOC<sub>s</sub> (24,048 records)  
 154 does not change the re-implementation result ( $R0 - A2 = +0.017$  pp,  $p = 0.67$ ), so the  
 155 difference is not a matter of data volume.

156 *Noise not seen in training.* When Gaussian noise is added only to the test  
 157 voltages of models trained on clean data, both neural models deteriorate strongly  
 158 ( $\sigma_{\text{noise}} = 0.5$  mV: R0 4.37, A2 4.60 pp), because difference- and spectrum-based  
 159 features computed on the smooth, 0.1-mV-resolution resampled records are sensitive  
 160 to noise absent from training. Models should therefore be trained on data from the  
 161 target sensor, or with matching noise augmentation, as in the matched-noise  
 162 experiment of Section 4.6.

## Supplementary Algorithm S1

**Algorithm S1.** Training and inference of the dual-branch SOH estimation model (handcrafted backbone + pulse-waveform complement).

**Input:** cleaned record set  $D = \{(V, x, y, \text{cell}, \text{rpt}, \text{soc}, \text{pulse})\}$  with 16,938 records of 64 cells;  $F = 5$  folds, split seed 42, initialisation seeds  $\{0, 1, 2\}$ ;  $K = 50$ , redundancy threshold 0.98,  $\lambda = 0.1$ .

**Output:** record-level test predictions and cell-macro errors for every fold and seed.

### Offline training

1: Remove cumulative cycle count, RPT index, coulomb-counted SOC and SOC deviation from the 150 candidates  $\rightarrow$  146 selectable features (Section 2.3.2)

2: Permute the 64 cells with seed 42 and divide them into five disjoint test folds (13/13/13/13/12 cells)

3: In each fold, draw validation cells from the non-test cells with the fold index as offset and stride 5

(train/validation/test = 40/11/13, 41/10/13, 41/10/13, 41/10/13, 42/10/12)

4: **for** outer fold  $f = 0, \dots, 4$  **do**

5:     Split  $D$  by cell into  $D_{tr}$ ,  $D_{va}$ ,  $D_{te}$ ; all RPTs and condition records of a cell stay together

6:     Compute Spearman  $\rho_j$  of the 146 candidates with SOH on  $D_{tr}$  only  $\triangleright$  Eq. (4)

7:     Select  $K$  features greedily by  $|\rho_j|$ , skipping rank correlation  $\geq 0.98$  with selected ones; fill to  $K$  if needed  $\triangleright$  Eq. (5)

8:     Fit medians, feature means/s.d., per-position waveform means/s.d., cycle-count and label statistics on  $D_{tr}$  only

9:     Transform  $D_{tr}$ ,  $D_{va}$ ,  $D_{te}$  with these statistics  $\triangleright$  Eqs. (6)–(10)

10:    **for** seed  $s \in \{0, 1, 2\}$  **do**

11:        Initialise  $E_h$ ,  $E_v$  and  $g$  with seed  $s$ ; AdamW (lr  $10^{-3}$ , weight decay  $10^{-4}$ ), batch 256,

ReduceLROnPlateau (patience 5, factor 0.5), gradient clipping 5, at most 100 epochs, early-stopping patience 15

12:        **for** each epoch and mini-batch **do**

13:            Enable gradients on  $\tau$ ; forward pass gives  $\hat{y}^* \triangleright$  Eq. (11)

14:            Compute  $\partial \hat{y}^* / \partial \tau$  by automatic differentiation (create\_graph = True)

15:             $\mathcal{L} \leftarrow \text{MSE}(\hat{y}^*, y^*) + \lambda \cdot \{[\partial \hat{y}^* / \partial \tau]_*^2\}$ ; back-propagate, clip, update  $\triangleright$  Eq. (13)

16:            After each epoch, compute the cell-macro RMSE on  $D_{va}$ ; keep the best weights; stop after 15 epochs without improvement

17:        Load the best-epoch weights and predict  $D_{te}$

### Online inference (one pulse record)

18: Reference the 101-point voltage to its first sample and standardise with the fold statistics  $\rightarrow \tilde{v} \triangleright$  Eqs. (6), (7)

19: Compute the  $K$  selected features, impute and standardise, append standardised  $\tau \rightarrow \mathbf{h} \triangleright$  Eqs. (8), (9)

20:  $\hat{y}^* \leftarrow g([E_v(\tilde{v}); E_h(\mathbf{h})])$ ; SOH  $\leftarrow 100(\sigma_y \hat{y}^* + \mu_y) \triangleright$  Eqs. (11), (10)

### Metrics and statistics

21: Average the six condition predictions of each cell–RPT, compute per-cell RMSE and the cell-macro mean  $\triangleright$  Eq. (14)

22: Paired differences: 5,000 fold-stratified cell bootstrap intervals; 5,000 two-sided paired sign-flip tests; Holm correction within families

## Supplementary Tables

**Supplementary Table S1.** Ageing protocols of the cycling-condition groups.

| Group | CC–CV charge rate (C) | CC discharge rate (C) | DOD (%) | Cells (IDs) |
|-------|-----------------------|-----------------------|---------|-------------|
| 1     | 2                     | 1.5                   | 60      | 6 (2–7)     |
| 2     | 2.42                  | 1.5                   | 60      | 6 (8–13)    |
| 3     | 2                     | 2                     | 60      | 6 (14–19)   |
| 4     | 2                     | 1.5                   | 80      | 6 (20–25)   |
| 5     | 2.83                  | 1.5                   | 60      | 6 (26–31)   |
| 6     | 2.42                  | 2                     | 60      | 6 (32–37)   |
| 7     | 2.42                  | 1.5                   | 80      | 6 (38–43)   |
| 8     | 3.25                  | 1.5                   | 60      | 6 (44–49)   |
| 9     | 2                     | 1.5                   | 100     | 6 (50–55)   |
| 10    | 2                     | 2                     | 80      | 6 (56–61)   |
| 11    | 2.83                  | 1.5                   | 80      | 4 (62–65)   |

From the cell test log in the publisher’s public repository [28]. Charging is CC–CV and discharging CC; rates refer to the nominal 1.2 Ah, so 2.42C, 2.83C and 3.25C correspond to 2.9 A, 3.4 A and 3.9 A.

**Supplementary Table S2.** Input information of the two branches.

| Input                                                   | Dimension | Waveform branch | Backbone branch         | Source                      |
|---------------------------------------------------------|-----------|-----------------|-------------------------|-----------------------------|
| First-sample-referenced voltage sequence                | 101       | ✓               | —                       | Single pulse                |
| Waveform-derived features (train-fold selected)         | 49        | —               | ✓                       | Single pulse                |
| Matched-condition DCIR (selected)                       | 1         | —               | ✓                       | Single pulse                |
| Standardised cumulative cycle count $\tau$              | 1         | —               | ✓                       | Cycle record                |
| Nominal SOC, pulse direction                            | —         | —               | Candidate, not selected | Test setting                |
| Coulomb-counted SOC, SOC deviation                      | —         | —               | Excluded                | Requires reference capacity |
| Other condition records, past capacities, cell identity | —         | —               | —                       | —                           |

✓, input enters the branch; —, not used or not applicable.

**Supplementary Table S3.** Ablation configurations and reference models.

| Category        | Configuration | Handcrafted backbone          | Waveform complement                    | $\tau$ | $\lambda$ | Trainings      |
|-----------------|---------------|-------------------------------|----------------------------------------|--------|-----------|----------------|
| Main            | R0            | MLP, 50 features              | PatchTST (9, 4)                        | ✓      | 0.1       | 15             |
| Encoder matrix  | 15 pairings   | MLP / ResNet / FT-Transformer | CNN / MLP / TCN / PW-Trans. / PatchTST | ✓      | 0.1       | 225 (incl. R0) |
| Branch ablation | A2            | MLP                           | —                                      | ✓      | 0.1       | 15             |

| Category         | Configuration                   | Handcrafted backbone     | Waveform complement                 | $\tau$ | $\lambda$ | Trainings                |
|------------------|---------------------------------|--------------------------|-------------------------------------|--------|-----------|--------------------------|
| Branch ablation  | A1                              | DCIR only (concatenated) | PatchTST (9, 4)                     | ✓      | 0.1       | 15                       |
| Factor ablation  | A3                              | MLP                      | PatchTST (9, 4)                     | ✓      | 0         | 15                       |
| Factor ablation  | A4                              | MLP                      | PatchTST (9, 4)                     | —      | 0         | 15                       |
| Patch parameters | P1, P2, P3                      | MLP                      | PatchTST (5, 2), (5, 4), (9, 2)     | ✓      | 0.1       | 45                       |
| Adaptation       | D1, D2, D3                      | MLP                      | Flatten head, + RevIN, raw voltage  | ✓      | 0.1       | 45                       |
| Feature count    | $K = 10, 20, 30, 75, 100, 144$  | MLP, $K$ features        | PatchTST (9, 4)                     | ✓      | 0.1       | 90 ( $K = 50$ reuses R0) |
| Published model  | Nowacki et al. [28], reproduced | —                        | Raw $\Delta V$ , $5 \times 100$ MLP | —      | —         | 30 models                |

211 Neural configurations use 5 folds  $\times$  3 seeds (465 trainings in total, plus 30 reproduced models).

212 **Supplementary Table S4.** Stratified waveform increment (R0 – A2, SOH pp).

| Stratum     | Level         | A2 RMSE | R0 RMSE | $\Delta$ RMSE | 95% CI           | Relative change | Improved cells |
|-------------|---------------|---------|---------|---------------|------------------|-----------------|----------------|
| True SOH    | <80%          | 2.404   | 1.991   | −0.413        | −0.460 to −0.363 | −17.2%          | 62/64          |
| True SOH    | 80–85%        | 1.742   | 1.601   | −0.141        | −0.200 to −0.082 | −8.1%           | 47/64          |
| True SOH    | 85–90%        | 1.974   | 1.861   | −0.113        | −0.151 to −0.074 | −5.7%           | 50/64          |
| True SOH    | 90–95%        | 1.150   | 1.123   | −0.027        | −0.046 to −0.009 | −2.3%           | 36/64          |
| True SOH    | $\geq 95\%$   | 0.956   | 0.808   | −0.148        | −0.163 to −0.134 | −15.5%          | 61/64          |
| Nominal SOC | 20%           | 1.546   | 1.452   | −0.094        | −0.116 to −0.072 | −6.1%           | 55/64          |
| Nominal SOC | 50%           | 1.550   | 1.414   | −0.136        | −0.161 to −0.111 | −8.8%           | 58/64          |
| Nominal SOC | 90%           | 1.579   | 1.474   | −0.104        | −0.127 to −0.080 | −6.6%           | 56/64          |
| Direction   | Charge        | 1.546   | 1.434   | −0.112        | −0.136 to −0.088 | −7.2%           | 55/64          |
| Direction   | Discharge     | 1.584   | 1.477   | −0.107        | −0.125 to −0.088 | −6.7%           | 57/64          |
| Condition   | 20% charge    | 1.533   | 1.431   | −0.102        | −0.128 to −0.076 | −6.7%           | 47/64          |
| Condition   | 20% discharge | 1.556   | 1.471   | −0.085        | −0.109 to −0.061 | −5.5%           | 49/64          |
| Condition   | 50% charge    | 1.512   | 1.380   | −0.132        | −0.161 to −0.104 | −8.7%           | 57/64          |
| Condition   | 50% discharge | 1.578   | 1.439   | −0.139        | −0.167 to −0.110 | −8.8%           | 54/64          |
| Condition   | 90% charge    | 1.567   | 1.458   | −0.109        | −0.141 to −0.078 | −7.0%           | 52/64          |
| Condition   | 90% discharge | 1.584   | 1.483   | −0.101        | −0.124 to −0.077 | −6.4%           | 54/64          |

213 True-SOH strata are formed at the cell–RPT level after averaging the six conditions, and each cell’s  
214 RMSE is computed within the stratum; SOC, direction and condition strata use single records.  
215 Predictions are averaged over the three seeds. Descriptive analysis without multiplicity correction.

**Supplementary Table S5.** Sensitivity to the number of handcrafted features (SOH pp).

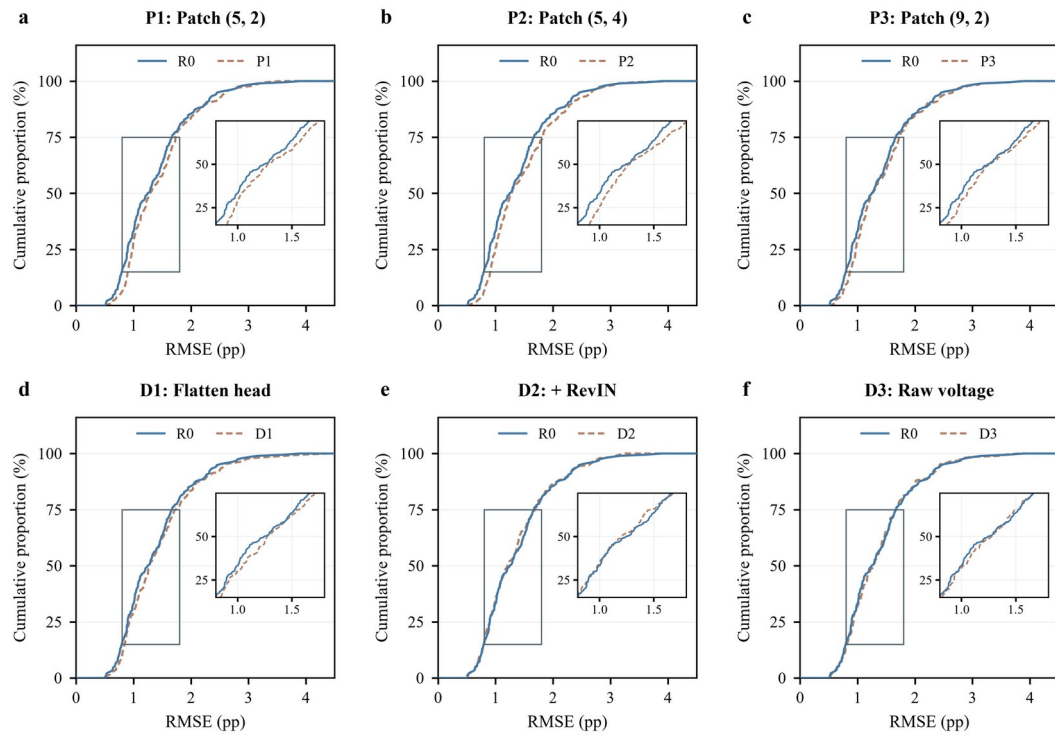
| $K$     | Backbone input | Parameters | RMSE  | MAE   | $\Delta$ RMSE | 95% CI          | Holm $p$ | Improved / worse cells |
|---------|----------------|------------|-------|-------|---------------|-----------------|----------|------------------------|
| 10      | 11             | 16,481     | 1.516 | 1.249 | +0.139        | 0.089–0.188     | 0.001    | 16/48                  |
| 20      | 21             | 17,121     | 1.499 | 1.226 | +0.122        | 0.077–0.169     | 0.001    | 14/50                  |
| 30      | 31             | 17,761     | 1.468 | 1.210 | +0.091        | 0.057–0.125     | 0.001    | 14/50                  |
| 50 (R0) | 51             | 19,041     | 1.377 | 1.141 | —             | —               | —        | —                      |
| 75      | 76             | 20,641     | 1.414 | 1.165 | +0.037        | 0.018–0.057     | 0.002    | 24/40                  |
| 100     | 101            | 22,241     | 1.418 | 1.176 | +0.041        | 0.021–0.062     | 0.002    | 15/49                  |
| 144     | 145            | 25,057     | 1.376 | 1.128 | −0.002        | −0.019 to 0.015 | 0.838    | 32/32                  |

$K$  excludes the separately appended cycle count, so the backbone input has  $K + 1$  dimensions.  $\Delta$ RMSE is the level minus  $K = 50$ ; Holm correction covers the six RMSE comparisons. Improved (worse) cells have lower (higher) error than  $K = 50$ .

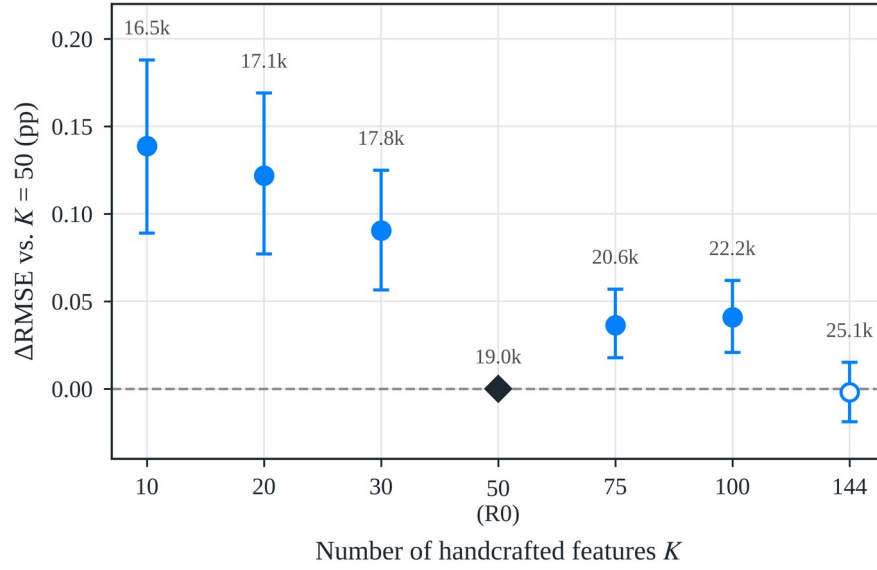
**Supplementary Table S6.** Results of the supplementary experiments (cell-macro RMSE, SOH pp).

| Scenario                                                               | Data | R0    | A2    | A1    | R0 – A2 | 95% CI           | $p$                 | Improved cells |
|------------------------------------------------------------------------|------|-------|-------|-------|---------|------------------|---------------------|----------------|
| Five-fold, reported (main text)                                        | LFP  | 1.377 | 1.500 | 1.494 | −0.123  | −0.141 to −0.103 | <0.001              | 58/64          |
| Five-fold, re-implementation                                           | LFP  | 1.427 | 1.474 | —     | −0.046  | −0.070 to −0.023 | <0.001              | 42/64          |
| Leave-one-group-out                                                    | LFP  | 1.560 | 1.594 | —     | −0.035  | −0.053 to −0.015 | 0.008               | 39/64          |
| Cycle count +10% (test)                                                | LFP  | 1.449 | 1.497 | —     | −0.048  | −0.072 to −0.024 | <0.001              | 43/64          |
| Cycle count −10% (test)                                                | LFP  | 1.477 | 1.518 | —     | −0.041  | −0.063 to −0.018 | 0.002               | 39/64          |
| Matched noise, $\sigma_{\text{noise}} = 1$ mV                          | LFP  | 1.620 | 1.678 | —     | −0.058  | −0.077 to −0.040 | <0.001              | 50/64          |
| Matched noise, $\sigma_{\text{noise}} = 2$ mV                          | LFP  | 1.715 | 1.760 | —     | −0.045  | −0.069 to −0.022 | 0.002               | 46/64          |
| Test-only noise, $\sigma_{\text{noise}} = 0.5$ mV <sup>a</sup>         | LFP  | 4.374 | 4.595 | —     | —       | —                | —                   | —              |
| Five-fold, original implementation <sup>b</sup>                        | NM C | 1.968 | 2.132 | 2.468 | −0.163  | to be added      | 0.0004 <sup>c</sup> | 39/44          |
| Five-fold, re-implementation                                           | NM C | 2.030 | 2.026 | 2.317 | +0.004  | −0.042 to 0.047  | 0.90                | 21/44          |
| Five-fold, re-implementation, backbone without absolute-level features | NM C | 1.940 | 2.098 | —     | −0.158  | −0.205 to −0.112 | <0.001              | 36/44          |

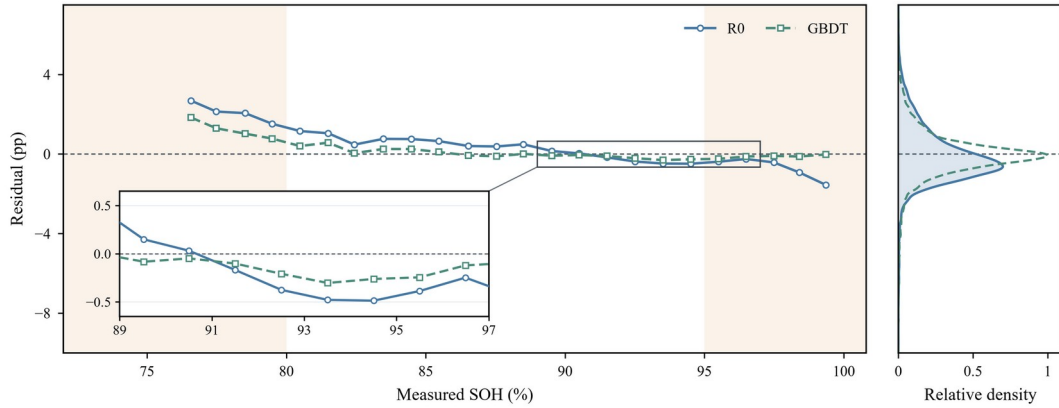
Paired comparisons use 5,000 fold- (or group-) stratified cell bootstraps and two-sided sign-flip tests, without multiplicity correction. <sup>a</sup> Models trained on clean data with noise added to the test voltages only. <sup>b</sup> Original implementation; DCIR added to the A1 input (Supplementary Note S4). <sup>c</sup> Holm-adjusted. Group-level leave-one-group-out errors are shown in Fig. 7a of the main text.



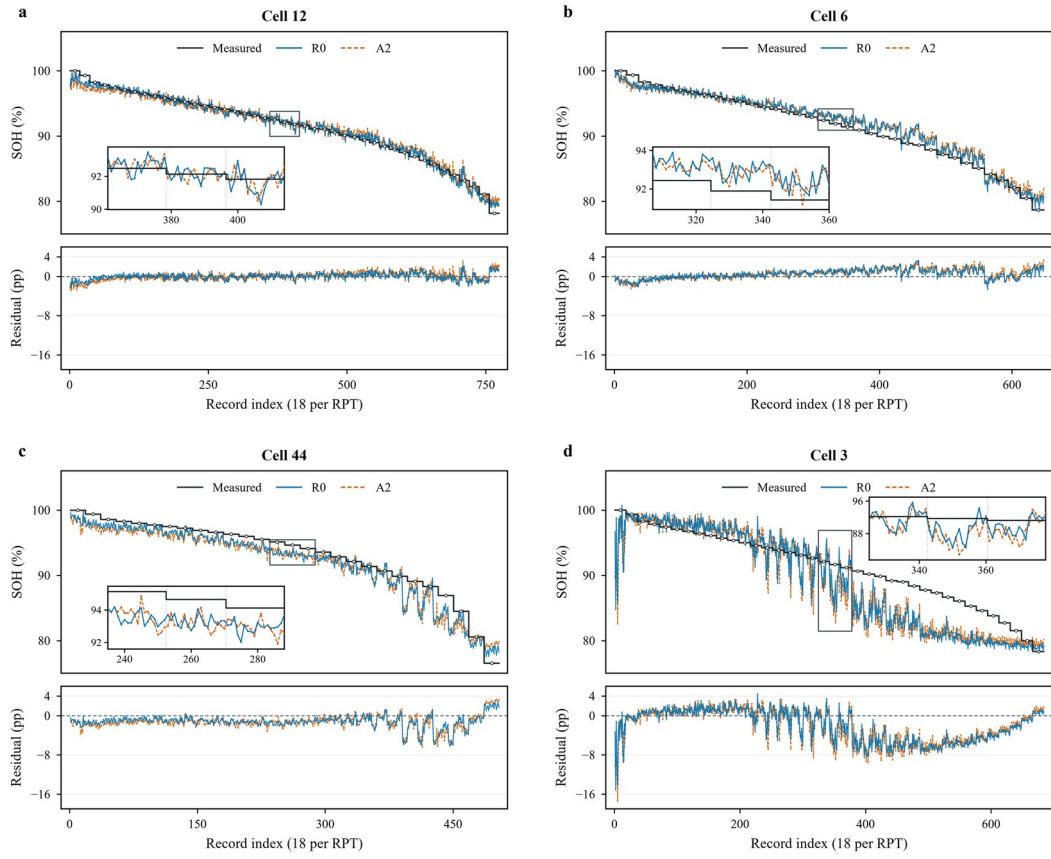
227  
 228 **Fig. S1.** Cumulative error distributions of the waveform-branch design variants. (a–c) Patch  
 229 configurations P1 (5, 2), P2 (5, 4) and P3 (9, 2) versus R0 (9, 4); (d–f) adaptation ablations  
 230 D1 (flatten head), D2 (with RevIN) and D3 (raw voltage). Solid blue, R0; dashed brown,  
 231 variant. Each curve contains the 192 cell  $\times$  seed RMSEs (not 192 independent cells); insets  
 232 magnify RMSE 0.8–1.8 pp and cumulative proportion 15–75%. Cell-level tests are given in  
 233 Table 8.



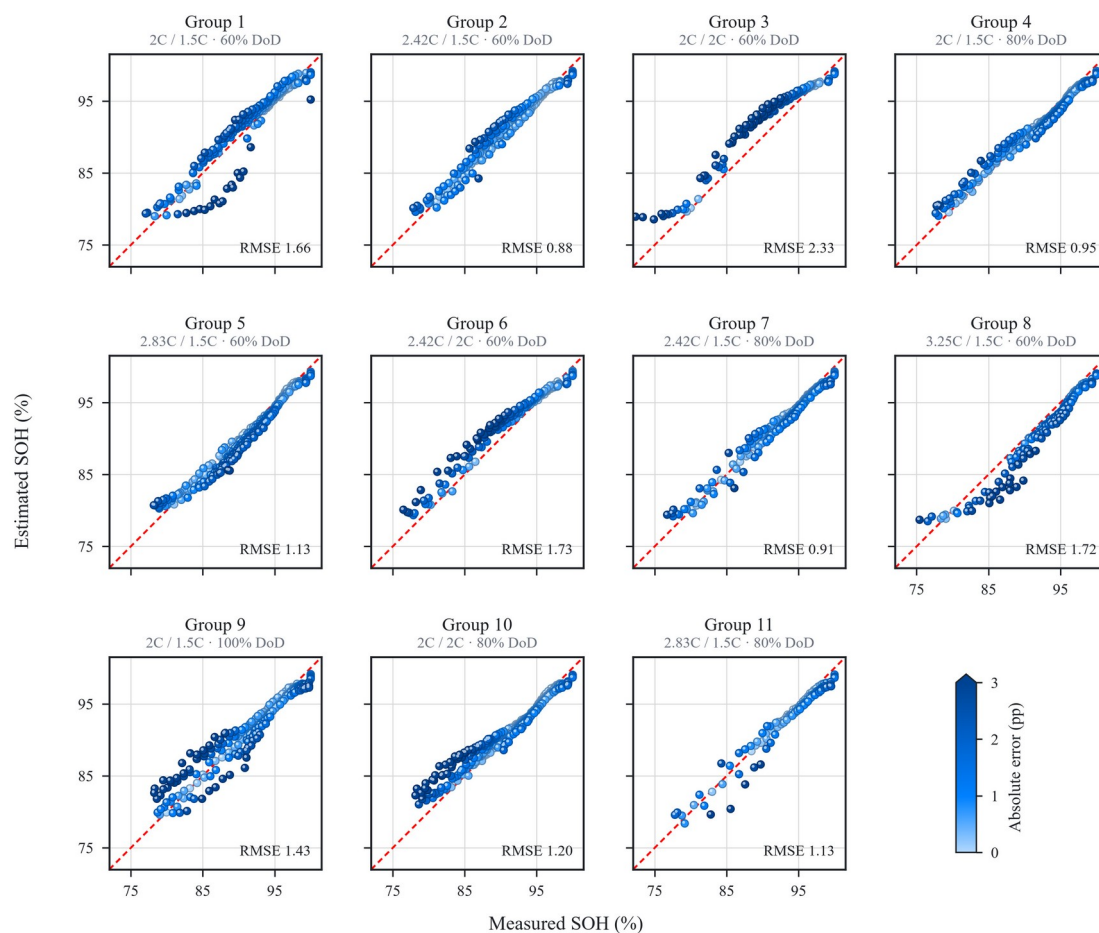
**Fig. S2.** Sensitivity to the number of handcrafted features  $K$ . Difference in cell-macro RMSE relative to  $K = 50$  with 95% fold-stratified cell bootstrap intervals; labels give the total parameter count. Hollow markers denote intervals that include zero.



**Fig. S3.** SOH-dependent mean bias and overall residual distribution of R0 (blue, solid) and GBDT (teal, dashed). Residuals are the seed-averaged prediction minus the measured SOH of each record. Left: 1-pp bins of measured SOH with at least 20 records; shading marks SOH < 80% and  $\geq 95\%$ ; the inset magnifies 89–97% SOH. Right: kernel density of all 16,938 residuals, normalised to the common maximum.



**Fig. S4.** Record-level predictions and residuals of representative cells 12, 6, 44 and 3 (ranks 1, 22, 43 and 64 by R0 error). Each RPT contributes 18 records (three SOCs  $\times$  two directions  $\times$  three seeds), so the horizontal axis is the record index rather than the cycle count; the grey staircase repeats the measured SOH of each RPT. Blue, R0; orange dashed, A2.



249  
 250 **Fig. S5.** SOH estimated by R0 versus measured SOH for the 11 cycling-condition groups.  
 251 Each point is one cell-RPT (predictions averaged over seeds and conditions, all from the  
 252 cell's test fold); colour denotes absolute error (saturating at 3 pp) and the red dashed line is  $y$   
 253  $= x$ . The number in each panel is the group's cell-macro RMSE following Eq. (14).